

UNIVERSITY OF CALIFORNIA, DAVIS
Department of Electrical and Computer Engineering

EEC284: Design and Optimization of Embedded Systems

Final Project: Fetal Heart Rate (FHR) Extraction from Multi-Channel mixed PPG

Submit solutions via a Canvas assignment. Please take note of the due date. The canvas assignment is configured to automatically stop accepting submissions at the due date.

The objective of this project is to develop an algorithm to extract Fetal Heart Rate (FHR) from dual-wavelength multi-channel transabdominal photoplethysmogram (PPG) data obtained from pregnant ewes. You are provided with the following:

- **Three sets of transabdominal PPG data** sampled at 80 Hz, high-pass filtered at 0.1 Hz. Each set corresponds to data collected in a portion of a controlled fetal blood desaturation experiment, aka hypoxic *round* or *round*, and includes 10 channels of data (five detectors, each measuring two distinct wavelengths of light).
- **Reference FHR signals** for each round, sampled at 1 Hz, serving as the ground truth.

Your task is to design an algorithm that leverages filtering, sensor fusion and other signal enhancement and processing techniques to extract the FHR from the multi-channel PPG signals as accurately as possible. Your approach should effectively enhance signal quality (e.g., minimize noise and artifacts) and combine information across the 10 channels. After processing the data, you will compare your estimated FHR with the reference FHR and evaluate the accuracy of your algorithm using specified performance metrics.

Submission Requirements:

1. **Code Implementation:** A well-documented code for your sensor fusion algorithm, ensuring reproducibility, clarity, and efficient execution. Comments should clearly explain the functionality of each section of your code. Please upload your code to a private github repository and submit the link. Please give both the instructor (ghiasi@gmail.com) and the TA (cfking@ucdavis.edu) view & download access to your repo.
2. **Report:** An IEEE-formatted report that provides a detailed explanation of your approach, including the following sections:
 - **Introduction:**
 - Briefly describe the problem and the objective of your project.
 - **Methods:**
 - Clearly explain the sensor fusion techniques you employed.
 - Justify your algorithm design and describe any preprocessing or signal enhancement methods applied.
 - Use figures and visuals to illustrates the performance of key steps of the method

- **Results and Discussion:**

- Present your findings, including a quantitative evaluation of your extracted FHR against the reference FHR using two performance metrics of Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- Include clear, well-labeled plots comparing your extracted FHR with both the reference FHR for each round of data, and prior art (i.e., the two published reference papers that were discussed in class).
- Provide a critical analysis of your algorithm's performance, discussing strengths, limitations, and potential sources of error.

- **Conclusions:**

- Summarize your key findings.
- Suggest potential further improvements for your sensor fusion approach. This component is meant to capture your thoughts on how the work could conceptually be extended, but does not need to be implemented as part of EEC284 project.

References

1. B. Kasap et al., "Multi-Detector Heart Rate Extraction Method for Transabdominal Fetal Pulse Oximetry," 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), 2021, pp. 1072-1075
2. B. Kasap, K. Vali, W. Qian, M. Saffarpour and S. Ghiasi, "KUBAI: Sensor Fusion for Non-Invasive Fetal Heart Rate Tracking," in IEEE Transactions on Biomedical Engineering, vol. 70, no. 7, pp. 2193-2202, July 2023